

SEMESTER END EXAMINATIONS – MARCH 2024

Program		B.E. – Artificial Intelligence and Data Science / Artificial Intelligence and Machine Learning	Semester	: III
Course Name		: Data Structures	Max. Marks	: 100
Course Code		: AD33 / AI33	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Explain dynamic memory allocation with suitable examples. CO1 (08)
 - Explain `strncpy()` and `strtok()` functions with suitable syntax and examples. CO1 (08)
 - Differentiate between structure and union. CO1 (04)
- Define a structure called *Student* with the member's *student_name*, *RollNo*, *admission_date* and *Dept*, where *admission_date* is a nested structure. Write a C program which uses functions to read the data of 10 students and display data department wise. CO1 (06)
 - Write a C function that extracts a substring of specified length from a string. The substring starts with the character specified by the user. CO1 (06)
 - Show the sparse matrix representation of the following matrix and write a c function to find the transpose of the same. CO1 (08)

0	0	14	0	0	7
0	-4	0	0	0	0
0	0	0	5	0	2
12	0	0	0	9	0
0	24	0	0	0	19

UNIT - II

- Apply infix to postfix conversion algorithm to convert following infix expressions to postfix expressions: CO2 (08)
 - $((A-(B+C))*D)/(E+F)$
 - $(A+B-C*(D-E)/F)-G$
 - Write C functions to perform add and delete operations on a circular queue using array implementation. CO2 (08)
 - Differentiate between stack and queue. CO2 (04)
- Explain the role of stack as a system stack with a neat diagram. CO2 (06)
 - Write a C program to evaluate the following postfix expression $7\ 2\ 3\ +\ -\ 3\ 8\ 2\ /\ * -$. Tabulate the input, stack contents and the output. CO2 (08)
 - Describe limitations of a linear queue with suitable examples. CO2 (06)

UNIT - III

- What is Doubly Linked List? Write an algorithm to insert and delete a node in Doubly Linked List. CO3 (06)
 - Illustrate the addition of following two polynomials represented using singly linked lists with suitable steps. CO3 (08)
$$A(x) = 8x^4 + 5x^3 - 5x^2 - 4$$

$$B(x) = 2x^5 - 4x^4 + 3x^3 + 12x$$

- ## UNIT- IV

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graph TD
 A((A)) --> B((B))
 A --> C((C))
 B --> D((D))
 B --> E((E))
 C --> F((F))
 C --> G((G))
 D --> H((H))
 D --> I((I))
 E --> J((J))

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- ## UNIT - V

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